

Exp. No 3 False Positive and True Incident Analysis

Aim : To analyze security alerts using Python and classify them as:

- True Positive (TP)
- False Positive (FP)
- False Negative (FN)
- True Negative (TN)

And to calculate:

- Precision
- Recall
- Accuracy
- Confusion Matrix

```
(unknown@kali) - [~/False_Positive_Lab]
$ python3 alert_analysis.py
===== FALSE POSITIVE & TRUE INCIDENT ANALYSIS =====

Dataset Loaded Successfully

  alert_id predicted_label actual_label
0         1           Attack      Attack
1         2           Attack      Normal
2         3           Normal      Attack
3         4           Attack      Attack
4         5           Normal      Normal
5         6           Attack      Attack
6         7           Normal      Normal
7         8           Attack      Normal
8         9           Normal      Attack
9        10           Normal      Normal

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Confusion Matrix Values:
True Positive (TP): 3
False Positive (FP): 2
False Negative (FN): 2
True Negative (TN): 3

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Performance Metrics:
Precision: 0.60
Recall: 0.60
Accuracy: 0.60

Analysis Completed Successfully.
Traceback (most recent call last):
  File "/home/unknown/False_Positive_Lab/alert_analysis.py", line 36, in <module>
    EOF
NameError: name 'EOF' is not defined
```